

INTRODUCTION

The use of computer technology has the greatest impact on the production system over the last decades. The old traditional ways of manual preparation in industrial design and manufacturing is time consuming and labour intensive. The trend in manufacturing today demands high quality products, low production cost and timely delivery of manufactured products which can be achieved by the use of computer technology. Development in the computer based industry has impact on the manufacturing process as well as the management and marketing components, therefore the modern day use of computer aided technology in the manufacturing industries will go long way in improving the productivity and product development

THE MEANING OF CAD

The term CAD (computer aided design) is defined as the use of computer system to assist the designer in creation, modification analysis or optimization of design. It can also be referred to as the use of computer software and related computer hardware that supplement or replaces the traditional hand tools in creating models and technical drawing. Basically, engineering drawing has been part and parcel of industry for many years. It is the bridge between engineering design and manufacturing process. Design information is easily communicated to manufacturers in the form of drawing prepared according to the defined drafting standard.

BRIEF HISTORY OF AUTOCAD

AutoCAD is one of the CAD system software introduced in December 1982 by Autodesk LTD in Sansalicia California of USA. It has become one of the a universal tool in the technical world today. The use of AutoCAD software has revolutionized the manner of drawing preparation. It also gives the basics for learning other CAD packages because of most the commands, terms and concepts are used universally.

ADVANTAGES OF CAD SYSTEM

There are many benefits that can be derived from the use of CAD system. Some of these advantages are:-

***Improved Quality**

The use of CAD allows designer to control the quality of the engineered product. It is very easy to investigate an error, diagnose the problem, and solve it using the CAD software before any prototype is made. This saves time and money.

***Productivity improvement in design**

CAD software allows team of designers to lower production costs, work faster and smarter and ultimately leads to quicker methods of project execution. Since designer can perform their work more efficiently and effectively, companies or industries are able to keep their teams small. This helps organizations to produce high quality, low cost products and lets them market their products faster.

*Customer desired modification made easy

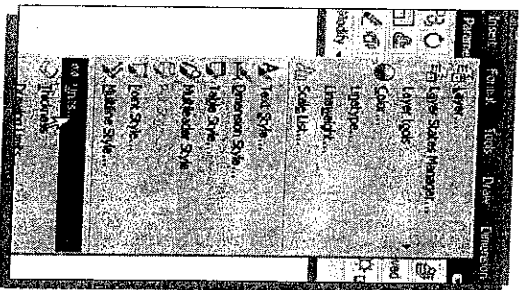
Customer modifications are easier to make. It eliminates tedious manual editing by providing a variety of editing tools. You can modify the existing object by mirroring, rotating, scaling, stretching, trimming or moving it to desired location. Changes can also be made on the object properties, such as line type, line weight, colour and layer at any time.

***Reduction of drafting labour**

The use of CAD software decreases design effort. The amount of effort needed for designing different models is reduced significantly because the software automates most of the task

***Saves time**

CAD design software saves designer time significantly. He can make better and more efficient designs in shorter time duration



***Improved accuracy**

The CAD design software provide an intrinsic capability for avoiding man made design, drafting and documentation errors. You have a tools to measure the precision, skill and accuracy level of the designs. Some of the tools errors that occurred because of manual designing is significantly reduced.

***Easy to share**

CAD software makes it easier for the designer to save the file and store it in a way that he use it time and again and send without any unwanted hassles.

DISADVANTAGES OF CAD SYSTEM

Though CAD system has a significant role to play in the manufacturing industries, yet it has some drawbacks which cannot be ignored. Some of the common disadvantages are:-

- *Work can be lost whenever the computer suddenly breakdown.
- *Designer work is prone to virus attack
- *Designed work could be easily hacked
- *It takes time to learn how to run the software.
- *High production or purchasing cost for CAD system cannot be avoided
- *Demand regular updating of software or operating system

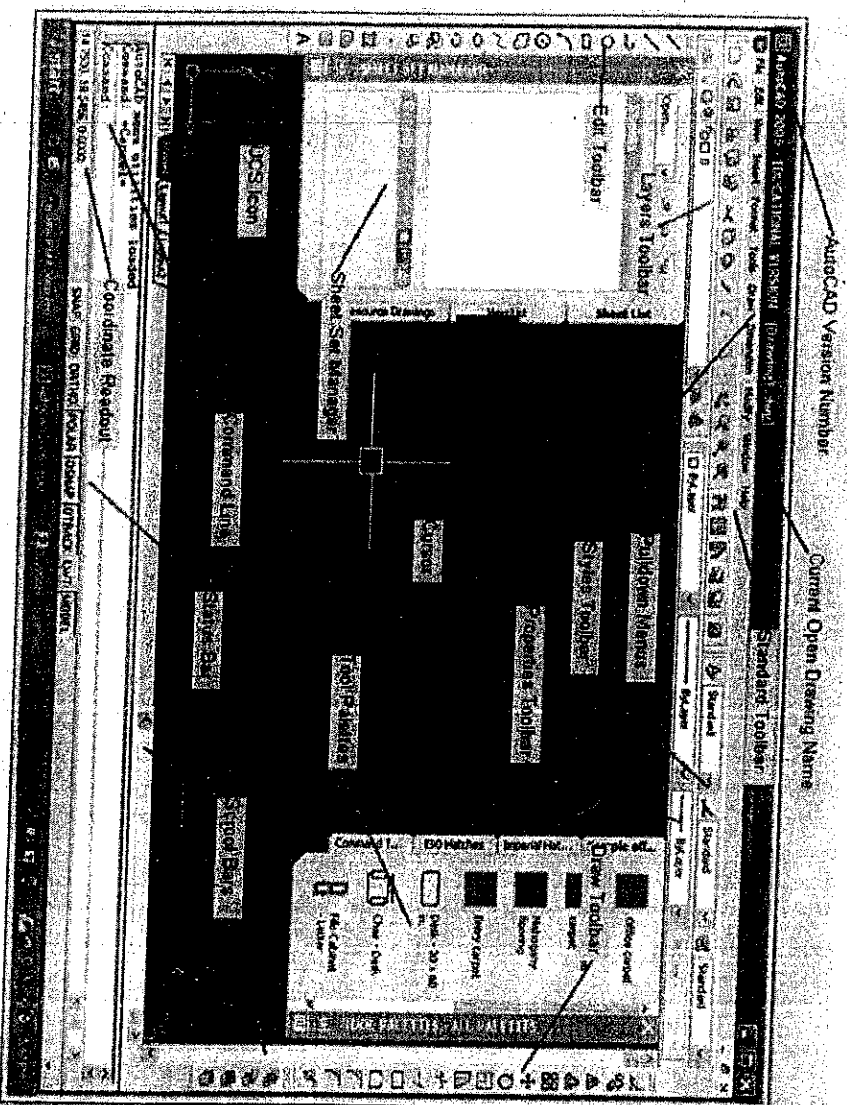
- * High cost of personnel training who will work on it.
- * Required less employment because of CAD system

APPLICATION OF CAD SOFTWARE

Engineers are using CAD Software in various industries for the following applications:-

- * To create concept or industrial designs and visualize them using realistic rendering software.
- * used to create three dimensional (3D) models of land, water or transportation features.
- * Assembly design
- * verification and validation of design (interference, gap analysis, tolerance stack up, draft and thickness analysis
- * CAD is also widely used to produce animation for special effects in movies, advertising and technical manuals often called DCC digital content creation
- * CAD system has also been major driving force for research in computational geometry and computer graphics.
- * Generating assembly and engineering drawings for manufacturing
- * CAD software enables engineers and architects to design, inspect and manage engineering projects within an integrated frame work.
- * CAD Software are used in construction industry for planning, designing and managing civil engineer in project.
- * CAD is important software extensively used in automotive industries, shipbuilding and aerospace industries and many more.

AUTOCAD SOFTWARE INTERFACE

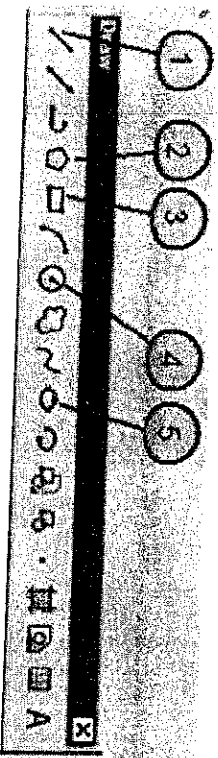


An interface is an application program that allows the user to interact with a computer. The goal of an effective interface is to enable the user experience easy and minimum effort in achieving the maximum desired outcome. AutoCAD Software has some features that make it easy for user to interact with the software. Some of the basic AutoCAD graphic window components are:-

- * AutoCAD version number- it shows what AutoCAD version you are running
- * Standard tool bar:-the tool bar for standard AutoCAD demand
- * Edit tool bar:-provide quick access to AutoCAD editing command
- * Status bar:-the status bar allows user to turn on and off AutoCAD settings
- * Layer tool bar:-provide access to AutoCAD layer commands
- * Styles toolbar:-gives access to AutoCAD styles commands
- * Properties toolbar:-provide access properties toolbar commands
- * Cursor:- used to point objects in AutoCAD drawing area
- * Draw toolbar:-gives access to AutoCAD drawing commands
- * Tool palettes:-access to tool palettes commands
- * Scroll bar:-used to scroll around drawing in drawing area
- * Command line:-allows user to type in AutoCAD command from keyboard
- * Coordinate read out:-used to keep track of cursor location in drawing area
- * Sheet manager:-gives access to sheet set manager commands
- * UCS (user coordinate system) icon:- this is a symbol with arrow at the bottom-left corner of the drawing area. The arrow point to the positive directions of the X and Y axes to help you keep your bearings.
- * Cross hair:-this is the two intersecting lines with a small box at their intersection. They show the location of mouse cursor in relation to other objects in your drawing.

DRAW COMMAND

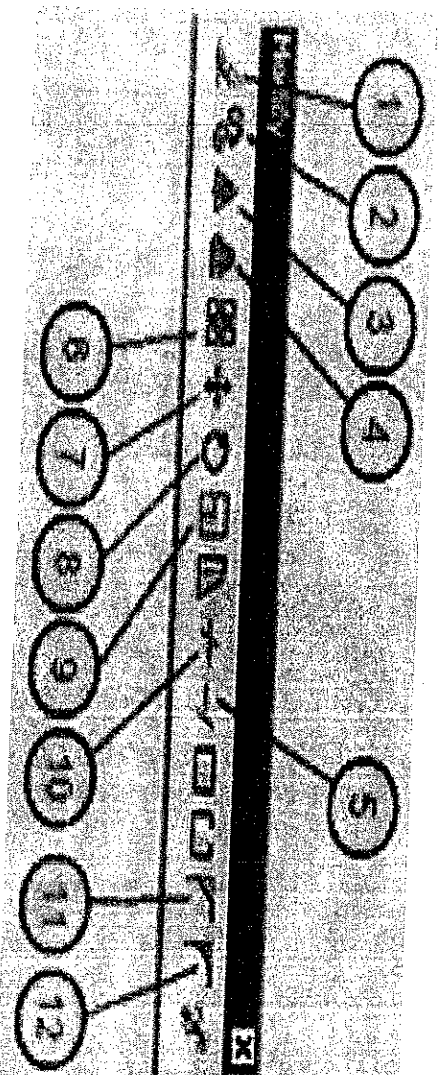
The draw commands are used to create objects such as rectangle, circle, lines, square and ellipse. Draw toolbar on the AutoCAD graphic window provides access to draw commands. Users can use draw pull-down menu or type the draw command in at the command line to activate the line command.



1. Line -used to draw lines in AutoCAD
2. Polygon – draws polygon with three or more sides
3. Rectangle –used to draw rectangles
4. Circle- this command is used to draw circles
5. Ellipse-used to draw ellipse

EDIT (MODIFY) COMMAND

Edit commands are used to modify the objects drawn in AutoCAD. The edit command can also be employed to moved, trim, rotate, offset, scaled or copied object drawn in AutoCAD.



1. Erase-this command is used to object drawn in the drawing area
2. Copy- used to copy one or more drawn in AutoCAD
3. Mirror- it is an edit command used to mirror an exact duplicate of an object

4. Offset - command used to offset one object from another to a distance you specify
5. Extend - used extend one line to another
6. Array- command used to make rectangular or polar array of an object
7. Move- an edit command used to move object around in the drawing area
8. Rotate -- used to rotate object around at a base point in the drawing area
9. Scale -- command use in AutoCAD to make object larger or small.
10. Trim -- used to trim object from another object
11. Chamfer -- command used to put chamfer between two lines
12. Fillet -- used to put a fillet between two lines at a specified radius